*Getting Started Guide*

# *Getting Started With C2000™ Real-Time Control Microcontrollers (MCUs)*

TEXAS INSTRUMENTS

## ABSTRACT

This guide is a valuable reference that contains all of the necessary information to get started with C2000™ real-time Microcontrollers (MCUs). This guide covers all aspects of development with C2000 devices from hardware to support resources. In addition to key reference documents, each section provides relevant links and resources to further expand on the information covered.

## Table of Contents



## List of Figures







入门指南

# *Getting Started With C2000TM Real-Time Control Microcontrollers (MCUs)*

TEXAS INSTRUMENTS

## 摘要

本指南是一本有价值的参考，包含了入门 C2000™ 实时微控制器 (MCU) 所需的所有信息。本指南涵盖了使用 C2000 器件进行开发的各个方面，从硬件到支持资源。除了关键参考文档之外，每个部分还提供了相关链接和资源，以便进一步扩展所涵盖的信息。

## 目录



## 文件列表gures

# Trademarks

C2000™, LaunchPad™, Code Composer Studio™, and InstaSPIN™ are trademarks of Texas Instruments.
MathWorks® is a registered trademark of The MathWorks, Inc.
All trademarks are the property of their respective owners.

# 1 Introduction

C2000 real-time MCUs are a portfolio of high-performance microcontrollers that are purpose-built to control power electronics and provide advanced digital signal processing for industrial and automotive applications. With more than 25 years of experience in developing microcontrollers optimized for simple real-time control, C2000 enables engineers to quickly create the world's most efficient power conversion and motor drive solutions.

C2000's product selection spans more than 100 devices to meet all levels of requirements. The available software packages make it easy for anyone to start their own software development. In addition, the reference designs and application kits enable highly optimized application specific solutions. This getting started guide is intended to provide all of the needed resources to take advantage and start development with C2000.

For a deeper look and understanding into the components that differentiate C2000 real-time microcontrollers as it pertains to Real-Time Control Systems, see the *The Essential Guide for Developing with C2000 Real-Time Microcontrollers* [Chinese].

# 2 Ecosystem

C2000's ecosystem is composed of various applications, products, hardware platforms, development tools, and software development kits. For more information on the specific sections of this ecosystem, see the respective chapters within this document. To learn about the essentials of C2000 devices and why C2000 is a leader in real-time control, watch the C2000™ Ecosystem 5-Minute Overview.

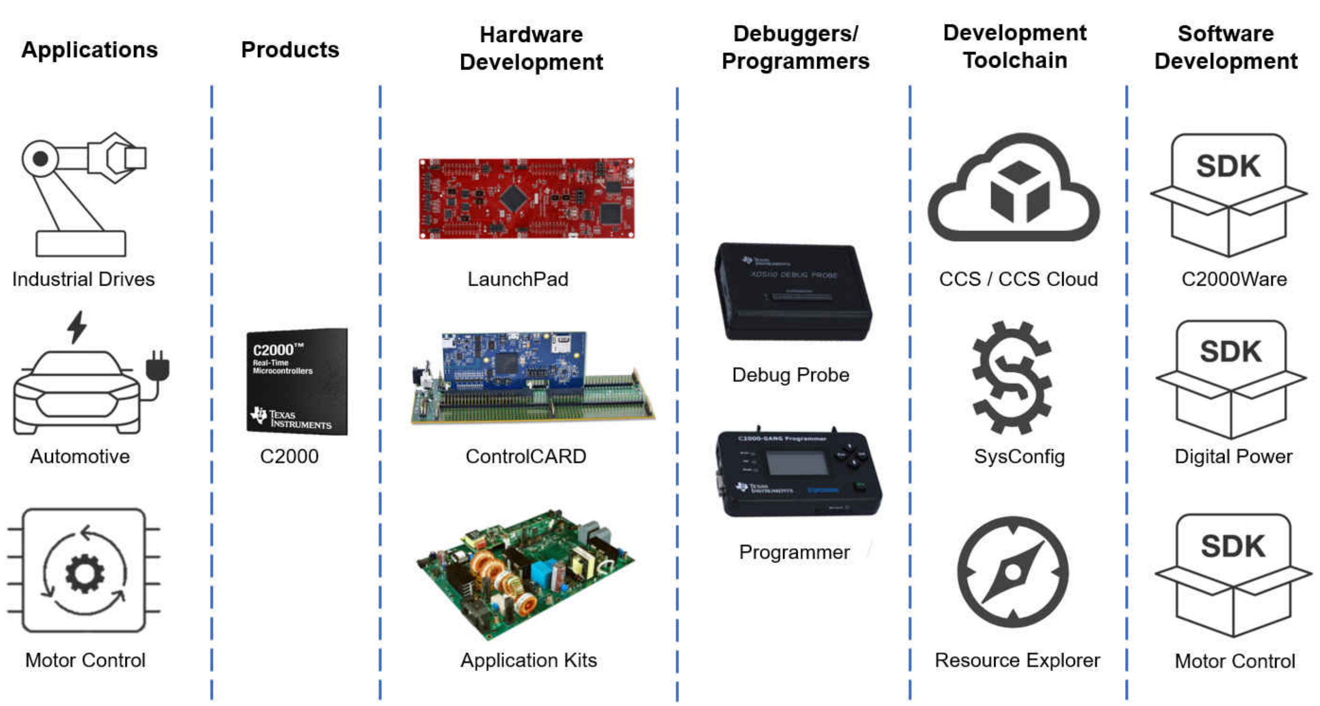


**Figure 2-1. C2000 Ecosystem Diagram**

The following sections discuss simplified ecosystem maps based on the familiarity level with C2000.







# 商标

C2000™、LaunchPad™、Code Composer Studio™ 和 InstaSPIN™ 是德州仪器的商标。MathWorks® 是 The MathWorks, Inc. 的注册商标。
所有商标均为其各自所有者的财产。

# 1 介绍

C2000 实时微控制器是一系列高性能微控制器，专为控制电力电子并为工业和汽车应用提供先进的数字信号处理而设计。凭借在为简单实时控制优化的微控制器方面超过 25 年的开发经验，C2000 使工程师能够快速创建全球最高效的电力转换和电机驱动解决方案。

C2000 的产品选择涵盖 100 多种器件，以满足各级需求。可用的软件包使任何人都能轻松开始他们的软件开发。此外，参考设计和应用套件使高度优化的特定应用解决方案成为可能。本入门指南旨在提供开始使用 C2000 并开展开发所需的所有资源。

For a deeper look and understanding into the components that differentiate C2000 real-time microcontrollers as it pertains to Real-Time Control Systems, see the *The Essential Guide for Developing with C2000 Real-Time Microcontrollers* [Chinese].

# 2 生态系统

C2000 的生态系统由各种应用、产品、硬件平台、开发工具和软件开发工具包组成。有关此生态系统特定部分的更多信息，请参阅本文档中的各相应章节。要了解 C2000 器件的基础知识以及为什么 C2000 在实时控制方面处于领先地位，请观看 C2000TM Ecosystem 5-Minute Overview.

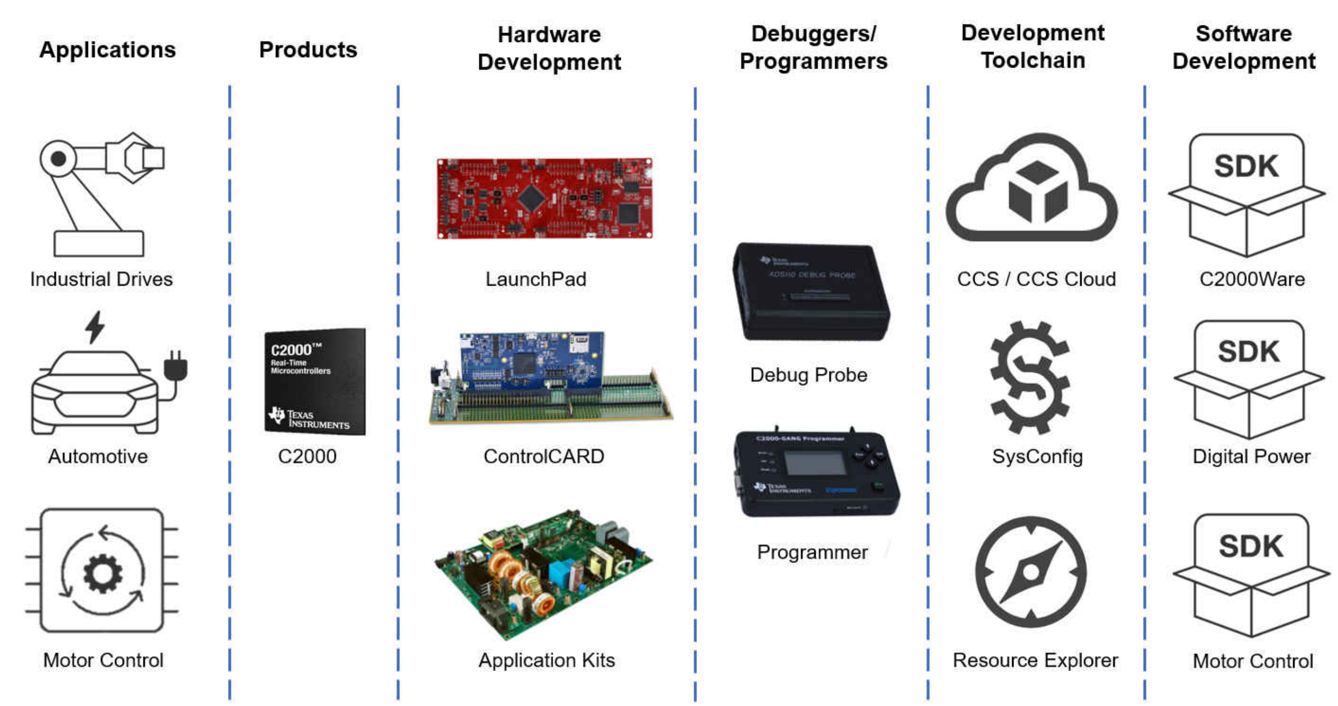


**图2-1 C2000 生态系统图**

下列各部分讨论了基于与 C2000 熟悉程度的简化生态系统图谱。

## 2.1 Entry Level

If you are new to C2000 and want to quickly explore device capabilities and features, then the best options to start with are any of the three low cost development platforms. These include a LaunchPad™, LaunchPad + BoosterPack, or ControlCARD + Docking Station. This will be coupled with Code Composer Studio (CCS) and the C2000Ware software development kit.

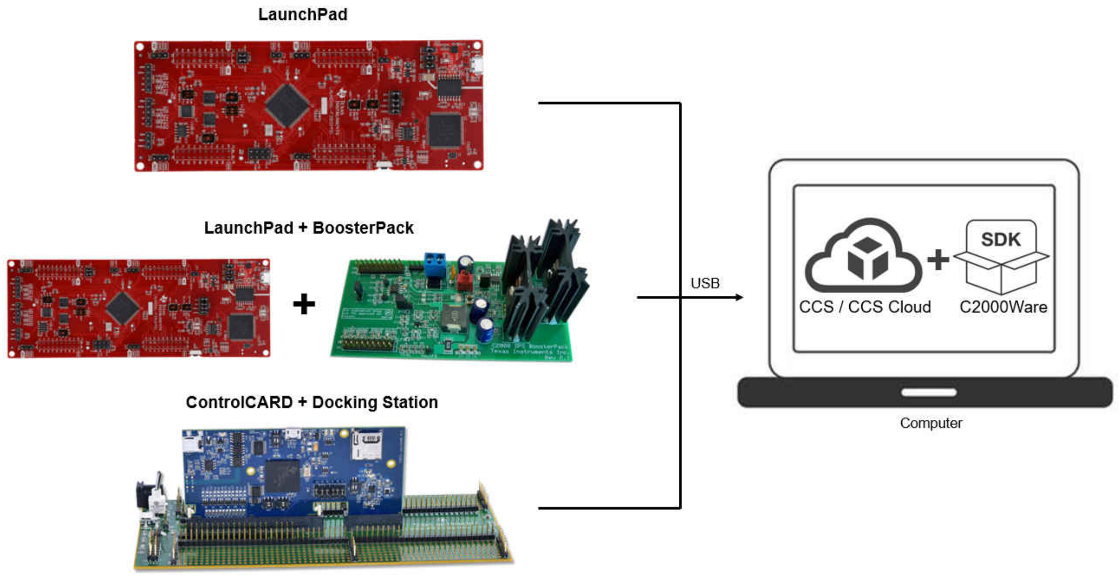


**Figure 2-2. C2000 Entry Level Ecosystem Diagram**

## 2.2 Intermediate Level

For those new to C2000, but exploring application specific development the best place to begin is with Section 3.5. These kits are categorized by power conversion and motor drive applications. Each category of application kits has its own software development kit (SDK), Digital Power SDK, and Motor Control SDK, respectively. These SDKs provide guided example code for each of the application kits enabling easy integration between hardware and software.

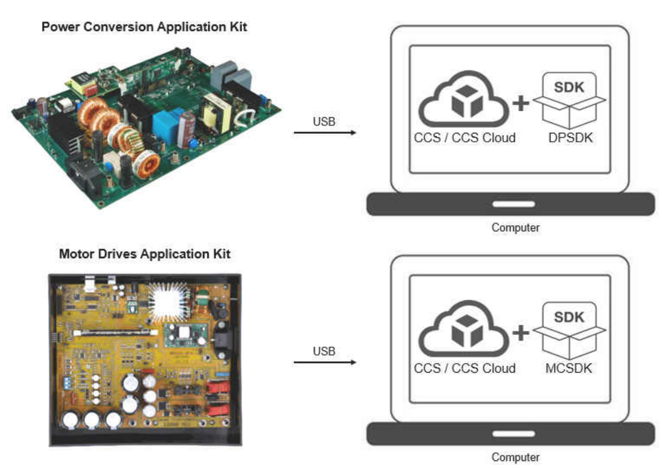


**Figure 2-3. C2000 Intermediate Level Ecosystem Diagram**







## 2.1 初级

如果您是 C2000 新手并且想快速探索器件能力和特性，那么最适合入门的选项是任何三种低成本开发平台之一。这些包括 LaunchPad™， LaunchPad + BoosterPack，或 ControlCARD + Docking Station。这将配合 Code Composer Studio (CCS) 和 C2000Ware 软件开发套件（SDK）。

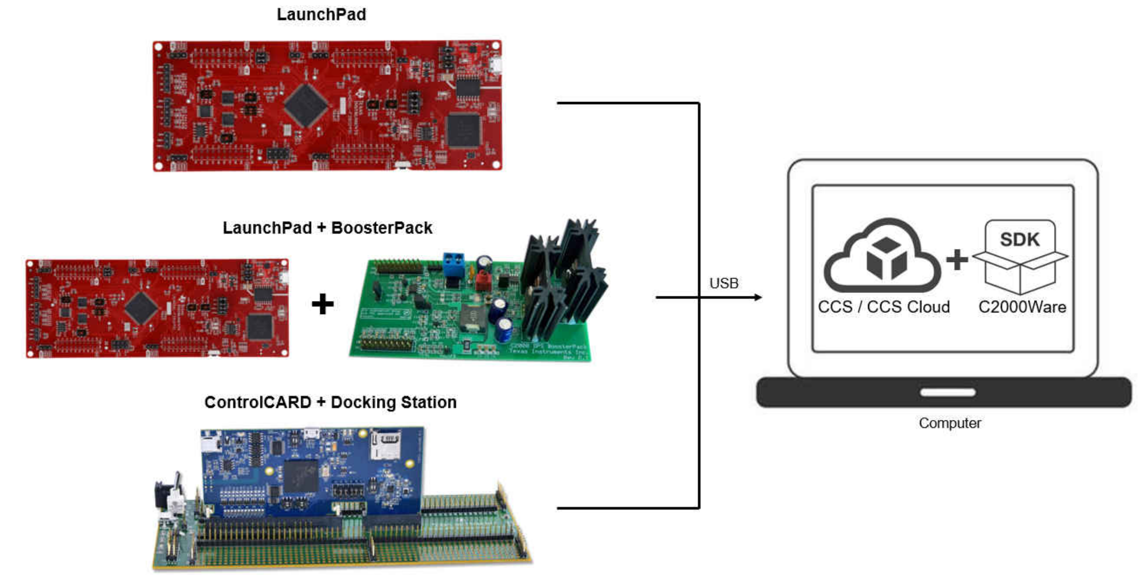


**图2-2。 C2000 入门级生态系统图**

## 2.2 中级

对于刚接触 C2000 但想探索特定应用开发的人来说，最好的起点是 第 3.5 节。这些套件按电力转换和电机驱动应用进行分类。每一类应用套件都有其自己的软件开发套件（SDK），数字电源 SDK 和 电机控制 SDK。这些 SDK 为每个应用套件提供了示例引导代码，便于硬件与软件之间的集成。

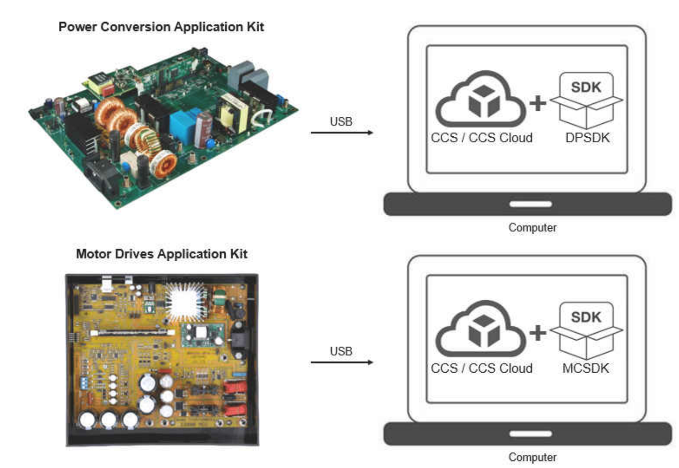


**图2-3。 C2000 中级生态系统图**

## 2.3 Advanced Level

If you are experienced with C2000 development, and are interested in quickly implementing a specific system then start with Section 3.6 and Section 3.7. These designs include all of the required files to jump start your development.

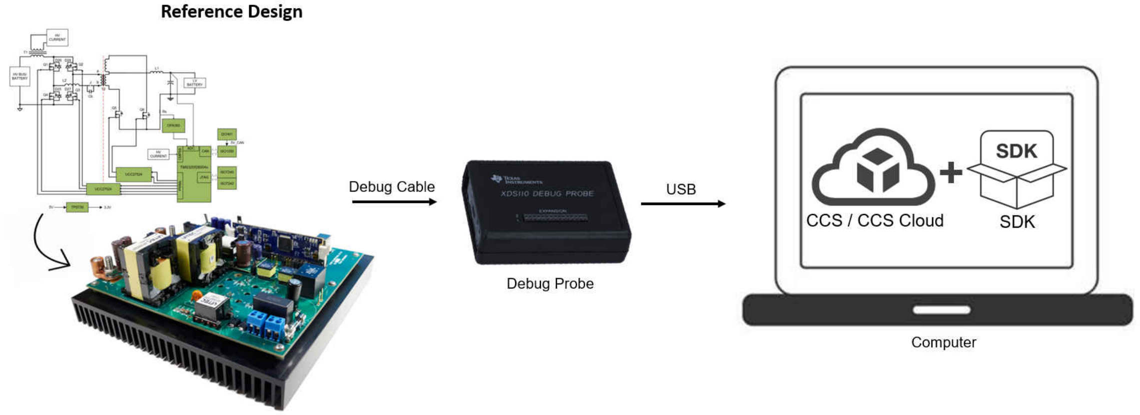


**Figure 2-4. C2000 Advanced Level Ecosystem Diagram**

# 3 Hardware Development

This section helps you understand C2000's entire hardware offering and how to quickly get started with hardware development.

## 3.1 LaunchPad

LaunchPads are low-cost development boards for the C2000 real-time microcontroller series of devices. Ideal for initial evaluation and prototyping, it provides a standardized and easy-to-use platform to develop an application.

For more information on available LaunchPads, see the LaunchPad development kits section of C2000's design and development page. Each LaunchPad comes with getting started materials that showcase how to setup your environment. LaunchPads also come with an onboard debug probe that allows for real-time debug and flash programming.

## 3.2 BoosterPack

BoosterPacks are pluggable add-on boards for the LaunchPad ecosystem that follow a pin-out standard created by Texas Instruments. The TI and third-party ecosystem of BoosterPacks greatly expands the peripherals and potential applications that you can explore with our various LaunchPads.

For C2000 BoosterPacks see the Full-featured evaluation modules section within C2000's design and development page.

To find the LaunchPads that are supported for each BoosterPack, see the 'Supported products' tab on the BoosterPack's page within TI.com.

## 3.3 ControlCARD

ControlCARDs are ideal to use for simple initial evaluation and system prototyping. ControlCARDs are complete board-level modules that utilize one of two standard form factors (100-pin DIMM or 180-pin HSEC) to provide a low-profile single-board controller solution.

ControlCARDs require either a Docking Station, a baseboard that powers the controlCARD and has a bread-board area for prototyping with an adequate DIMM or HSEC connection, or a compatible kit. For a complete package option, see Section 3.4 and Section 3.5.







## 2.3 高级

如果你有 C2000 开发经验，并且希望快速实现特定系统，那么请从 第3.6节 和 第3.7节 开始。这些设计包含了启动开发所需的所有文件。

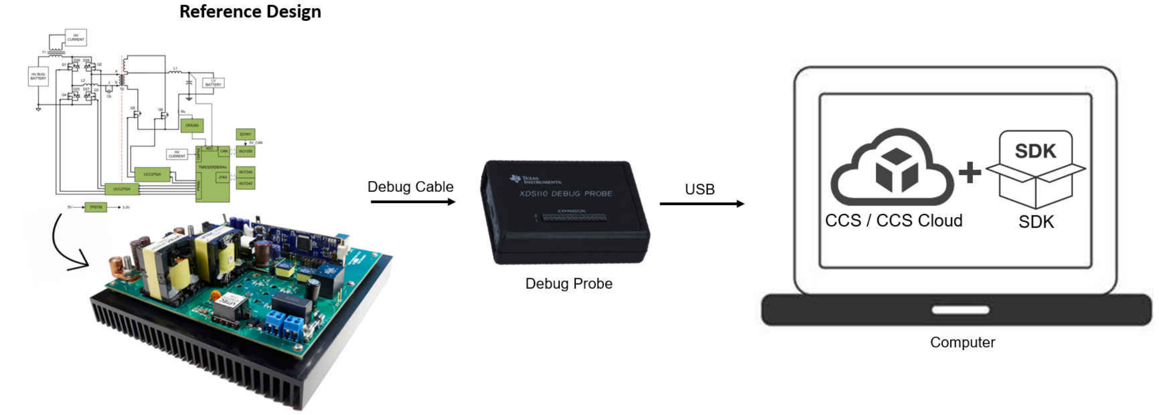


**图2-4。C2000 高级生态系统图**

# 3 硬件开发

本部分帮助您了解 C2000 的全部硬件产品，以及如何快速开始硬件开发。

## 3.1 LaunchPad

LaunchPad 开发板是面向 C2000 实时微控制器系列器件的低成本开发板。非常适合初始评估和原型开发，提供了一个标准化且易于使用的平台来开发应用。

有关可用 LaunchPads 的更多信息，请参见 LaunchPad development kits 部分，位于 C2000 的 design and development 页面。每个 LaunchPad 都附带展示如何设置您的环境的入门材料。LaunchPads 还配备了一个板载调试探针，可用于实时调试和 Flash Programming。

## 3.2 BoosterPack

BoosterPacks 是面向 LaunchPad 生态系统的可插拔扩展板，遵循由 德州仪器 (Texas Instruments) 制定的引脚排列标准。TI 和第三方的 BoosterPack 生态系统大大扩展了各种 LaunchPads 可探索的外设和潜在应用。

有关 C2000 BoosterPacks 的信息，请参见 Full-featured evaluation modules 部分，位于 C2000 的 design and development 页面。

要查找每个 BoosterPack 支持哪些 LaunchPads，请参见 TI.com 上 BoosterPack 页面中的“Supported products”选项卡。

## 3.3 ControlCARD

ControlCARD 板卡非常适合用于简单的初始评估和系统原型开发。ControlCARD 板卡是完整的板级模块，采用两种标准外形之一（100 针 DIMM 或 180 针 HSEC），以提供低剖面单板控制器解决方案。

ControlCARD 板卡需要对接站、为 ControlCARD 供电并具有用于原型开发的面包板区域且带有合适 DIMM 或 HSEC 连接的主板，或兼容的套件。有关完整套装选项，请参见 第 3.4 节 和 第 3.5 节。

For more information on available ControlCARDs, see the Full-featured evaluation modules section within C2000's design and development page. Each ControlCARD's page contains all relevant documentation to help you get started on your development.

### 3.4 Experimenter Kit

Experimenter kits contain both a controlCARD and corresponding Docking Station (100-pin DIMM or 180-pin HSEC). The Docking Station provides power to the controlCARD and has a bread-board area for prototyping. Access to the key device signals are easily available using a series of header pins.

To find the appropriate baseboard, see the description on the product page for the ControlCARD.

### 3.5 Application Kit

Applications can be evaluated through BoosterPack modules paired with LaunchPads, experimenter kits, or in some cases EVMs with the MCU soldered directly to the board. Some application kits are aimed at experimentation and concept education, while others provide more real-world power levels similar to the end application. All application kits contain convenient software examples, simple user's guides, and a hardware development package to enable rapid customization for your product.

For a full list of application kits, see the Motor Control SDK and Digital Power SDK.

### 3.6 Reference Designs

The TI Reference Design Library is a robust reference design library spanning analog, embedded processor, and connectivity. Created by TI experts to help you jump start your system design, all reference designs include schematic or block diagrams, BOMs, and design files to speed your time to market.

Search and download C2000 Reference Designs

### 3.7 Custom Design

A great starting reference for a custom board design is the design files of C2000 hardware platforms, whether that be a reference design, ControlCARD, or LaunchPad platform. For your hardware platform of interest, the schematic, BOM, and other design files can be found on the web page.

Below are various documentation resources to help you get started with a custom design:

- Hardware Design Guide for F2800x C2000™ Real-Time MCU Series provides an overview of system level hardware design as well as information on how to transition from schematic design to board layout.
- *How to Maximize GPIO Usage in C2000™ Devices* lists several suggestions on maximizing the GPIO resources on the device to limit the need for IO expanders.
- *An Introduction to IBIS (I/O Buffer Information Specification) Modeling* discusses various aspects of IBIS including its history, advantages, compatibility, model generation flow, data requirements in modeling the input/ output structures, and future trends.

**Note**

The *Design Tools and Simulation* section of the design and development tab of each device's product folder offer various downloadable models. These models include I/O Buffer Information Specification (IBIS) Models and Boundary-Scan Description Language (BSDL) Modules.






有关可用 ControlCARD 板卡的更多信息，请参见 全功能评估模块 部分，该部分位于 C2000 的设计与开发页面。每个 ControlCARD 的页面均包含帮助您开始开发的所有相关文档。

### 3.4 Experimenter Kit

Experimenter kits 包含 controlCARD 和相应的 Docking Station（100-pin DIMM 或 180-pin HSEC）。Docking Station 为 controlCARD 提供电源，并且具有用于原型开发的面包板区域。通过一系列引脚排可以方便地访问关键器件信号。

要查找合适的主板，请参见 ControlCARD 的产品页面上的描述。

### 3.5 Application Kit

应用可通过与 LaunchPads 配对的 BoosterPack 模块、experimenter kits，或在某些情况下将 MCU 直接焊接到板上的 EVMs 来评估。有些 application kits 面向实验和概念教育，而其他套件提供更接近实际应用的功率等级。所有 application kits 都包含便捷的软件示例、简明的用户指南和硬件开发包，以便快速为您的产品进行定制。

For a full list of application kits, see the 电机控制 SDK and 数字电源 SDK。

### 3.6 Reference Designs

TI 参考设计库是一个涵盖模拟、嵌入式处理器和连接性的强大参考设计库。由 TI 专家创建以帮助您快速启动系统设计，所有参考设计均包含原理图或框图、物料清单 (BOM) 和设计文件，以加速产品上市时间。

Search and download C2000 参考设计

### 3.7 Custom Design

定制电路板设计的一个很好的起点是 C2000 硬件平台的设计文件，无论是参考设计、ControlCARD 还是 LaunchPad 平台。对于您感兴趣的硬件平台，其原理图、物料清单 (BOM) 和其他设计文件都可以在网页上找到。

以下是一些帮助您开始定制设计的各种文档资源：

- F2800x C2000TM 实时微控制器系列硬件设计指南 提供了系统级硬件设计概述以及有关如何从原理图设计过渡到电路板布局的信息。
- 如何在 C2000TM 器件中最大化 GPIO 使用 列出了一些建议，以最大化器件上的 GPIO 资源，从而减少对 IO 扩展器的需求。
- IBIS（I/O 缓冲区信息规范）建模简介 讨论了 IBIS 的各个方面，包括其历史、优势、兼容性、模型生成流程、建模输入/输出结构时的数据要求以及未来趋势。

**Note**

The 设计工具与仿真 部分位于每个器件的产品资料夹的设计与开发选项卡中，提供各种可下载模型。这些模型包括 I/O 缓冲器信息规范（IBIS）模型和 边界扫描描述语言（BSDL）模块。

# 4 Software Development

This section helps you understand C2000's entire software offering. The information covered in this section can also be accessed through the C2000™ Software Guide.

## 4.1 Guides

For relevant C2000 software collateral, see the C2000 software reference guides.

## 4.2 C2000Ware

C2000Ware is the core software development kit for C2000. This provides a cohesive set of low level drivers, math and DSP related development software and documentation designed to minimize software development time. C2000Ware also provides a Software Diagnostic Library that can be used for Functional Safety software development.

**Downloads:**

Standalone version: C2000WARE | Online version: C2000Ware TIREX

**Resources:**

- Getting Started with CCS and C2000Ware is a 10-minute video that provides an overview of CCS and C2000ware. It also explains how to import an example and run it on hardware.
- Introduction to C2000Ware is a quick video overview of the C2000Ware development package.
- *C2000Ware Quick Start Guide* explains C2000's package structure, support, and how to quickly access software examples through Code Composer Studio.

## 4.3 Motor Control Software Development Kit (MCSDK)

The MCSDK provides a cohesive set of software infrastructure, tools, and documentation designed to minimize C2000 MCU-based motor control system and software development time.

**Downloads:**

Standalone version: C2000WARE-MOTORCONTROL-SDK | Online version: MCSDK_TIREX

To view all of the solutions + evaluation modules supported in the most current SDK release, click on the 'View all' option within the features section of the MCSDK page.

**Resources:**

- *C2000Ware MotorControl SDK Getting Started Guide* explains MCSDK's package structure, support, and how to access software examples through Code Composer Studio

## 4.4 Digital Power Software Development Kit (DPSDK)

The DPSDK provides a cohesive set of software infrastructure, tools, and documentation designed to minimize C2000 MCU-based digital power system and software development time.

**Downloads:**

Standalone version: C2000WARE-DIGITALPOWER-SDK | Online version: DPSDK TIREX

Within the C2000 software guide you can view all of the solutions + evaluation modules supported in the most current DPSDK release.

**Resources:**

- *C2000Ware DigitalPower SDK Getting Started Guide* explains DPSDK's package structure, support, and how to access software examples through Code Composer Studio







# 4 软件开发

本部分可帮助您了解 C2000™ 的全部软件产品。此部分涵盖的信息也可以通过 C2000TM Software Guide 获取。

## 4.1 指南

有关相关的 C2000 软件材料，请参阅 C2000 软件参考指南。

## 4.2 C2000Ware

C2000Ware 是 C2000 的核心软件开发套件（SDK）。它提供了一套完整的底层驱动程序、数学和数字信号处理相关的开发软件以及旨在最小化软件开发时间的文档。C2000Ware 还提供了可用于功能安全软件开发的软件诊断库。

**下载：**

Standalone version: C2000WARE | Online version: C2000Ware TIREX

**资源：**

- Getting Started with CCS and C2000Ware 是一个 10 分钟视频，提供了 CCS 和 C2000Ware 的概述。它还解释了如何导入示例并在硬件上运行它。
- Introduction to C2000Ware 是一个关于 C2000Ware 开发包的快速视频概述。
- *C2000Ware Quick Start Guide* 介绍了 C2000 的包结构、支持以及如何快速访问通过 Code Composer Studio 访问软件示例。

## 4.3 电机控制软件开发套件 (MCSDK)

MCSDK 提供了一套连贯的软件基础设施、工具和文档，旨在尽量缩短基于 C2000 MCU 的电机控制系统和软件的开发时间。

**下载：**

Standalone version: C2000WARE-MOTORCONTROL-SDK | 在线版本: MCSDK _TIREX

To view all of the solutions + evaluation modules supported in the most current SDK release, click on the 'View all' option within the features section of the MCSDK page.

**资源：**

- C2000Ware 电机控制 SDK 入门指南 说明 MCSDK 的包结构、支持以及如何通过 Code Composer Studio 访问软件示例

## 4.4 数字电源软件开发套件 (DPSDK)

DPSDK 提供了一套完整的软件基础设施、工具和文档，旨在最大限度地减少基于 C2000 MCU 的数字电源系统和软件的开发时间。

**下载：**

Standalone version: C2000WARE-DIGITALPOWER-SDK | Online version: DPSDK TIREX

在C2000 软件指南中，您可以查看在最新 DPSDK 发布中支持的所有解决方案 + 评估模块。

**资源：**

- C2000Ware DigitalPower SDK 入门指南 解释了 DPSDK 的包结构、支持以及如何通过 Code Composer Studio 访问软件示例

# 5 Programmers and Debuggers

## 5.1 Flash Programming

TI and multiple third parties offer several hardware and software solutions for performing both in-system and off-board programming of C2000 devices.

**Resources:**

- C2000 3P Search Tool contains a full list of the third party options avaliable for production programming.
- Flash: Frequently Asked Questions has answers to frequently asked questions about flash and flash programming.
- *Serial Flash Programming of C2000 Microcontrollers* describes one possible implementation to program the target device's on-chip flash memory.

## 5.2 Debug Probes

JTAG debug probes allow you to program the memories and communicate with a C2000 real-time MCU during development. While almost all C2000 tools include JTAG emulation on the LaunchPad, controlCARD, or application kit, once you build your own board you will need an external debug probe.

**Resources:**

- The 'Emulation/JTAG' section of the datasheet contains information on the various JTAG signals as well as the physical connection implementation.
- The Debug Probes section of C2000's design and development page has a list of the debug probes applicable to the C2000 family of products.
- *C2000 MCU JTAG Connectivity Debug* provides a brief overview of JTAG implementation and steps to resolve common JTAG connectivity errors when using Code Composer Studio software.

# 6 Development Toolchain

## 6.1 Code Composer Studio (CCS)

Code Composer Studio™ is an integrated development environment (IDE) that supports TI's Microcontroller and Embedded Processors portfolio. Code Composer Studio comprises a suite of simple tools used to develop and debug embedded applications. It includes an optimizing C/C++ compiler, source code editor, project build environment, debugger, profiler, and many other features. The IDE provides a single interface, taking the user through each step of the application development flow. Familiar tools and interfaces allow users to get started faster than ever before.

**Downloads:**

Standalone version: CCSTUDIO | Online version CCS Cloud

**Resources:**

- *Development Tool Versions for C2000 Support* lists the CCS and compiler versions required to develop applications targeting different C2000 features and devices.
- *Code Composer Stuido User's Guide* and the CCS Resources page explain the features and capabilities of the Code Composer Studio IDE.

## 6.2 SysConfig System Configuration Tool (SYSCONFIG)

C2000 SysConfig is a tool that exists integrated in CCS or as a standalone program that allows the user to generate C header and code files using a GUI. This interface creates a layer above the complex driverlib functions and simplifies the development process significantly. The generated code can be used with SDK examples or to configure custom software. The SysConfig tool also contains simple graphical utilities for configuring pins, peripherals, subsystems, and other components. SysConfig can help you quickly manage, expose, and resolve conflicts visually so that you have more time to create differentiated applications.






# 5 程序员和调试器

## 5.1 闪存编程

TI 和多家第三方提供了多种用于对 C2000 器件执行系统内编程和板外编程的硬件和软件解决方案。

**资源：**

- C2000 3P Search Tool 包含可用于生产编程的第三方选项的完整列表。
- Flash: Frequently Asked Questions 对关于 flash 和闪存编程的常见问题提供了解答。
- *Serial Flash Programming of C2000 Microcontrollers* 描述了用于对目标器件的片上 Flash 存储器进行编程的一种可能实现。

## 5.2 调试探针

JTAG 调试探针可让您在开发过程中对存储器进行编程并与 C2000 实时微控制器通信。虽然几乎所有的 C2000 工具在 LaunchPad 开发板、ControlCARD 或应用套件上都包含 JTAG 仿真，但一旦您自行构建电路板，就需要一个外部调试探针。

**资源：**

- 数据手册的“仿真/JTAG”部分包含有关各种 JTAG 信号以及物理连接实现的信息。
- The 调试探针 部分的 C2000 产品系列 的 设计与开发 页面 列出了适用于 C2000 的调试探针。
- *C2000 MCU JTAG Connectivity Debug* 提供了关于 JTAG 实现的简要概述以及在使用 Code Composer Studio 软件时解决常见 JTAG 连接错误的步骤。

# 6 开发工具链

## 6.1 Code Composer Studio (CCS)

Code Composer Studio™ 是一个支持德州仪器Microcontroller和Embedded Processors产品组合的集成开发环境。Code Composer Studio包含一套用于开发和调试嵌入式应用的简单工具。它包括一个优化的C/C++ 编译器、源代码编辑器、项目构建环境、调试器、分析器以及许多其他特性。该集成开发环境提供了一个单一界面，引导用户完成应用开发流程的每一步。熟悉的工具和界面使用户比以往更快入门。

**下载**

Standalone version: CCSTUDIO | Online version CCS 云

**资源：**

- *Development Tool Versions for C2000 Support* 列出开发针对不同 C2000 特性和器件的应用所需的 CCS 和编译器版本。
- *Code Composer Stuido User's Guide* 和 CCS Resources 页面解释了 Code Composer Studio IDE 的特性和功能。

## 6.2 SysConfig 系统配置工具 (SYSCONFIG)

C2000 SysConfig 是一个集成在 CCS 中或作为独立程序存在的工具，允许用户使用 GUI 生成 C 头文件 和 代码文件。该界面在复杂的 driverlib 函数之上创建了一层抽象，并显著简化了开发过程。生成的代码可以与 SDK 示例一起使用或用于配置自定义软件。SysConfig 工具还包含用于配置 引脚、外设、子系统 和其他 组件 的简单图形实用程序。SysConfig 可以帮助您快速以可视化方式管理、暴露并解决冲突，从而让您有更多时间创建差异化的 应用。

**Downloads:**

C2000 SysConfig is delivered through C2000Ware and can be used within CCS. To access SysConfig, either start from an existing C2000 SysConfig-based driverlib project (these projects will contain a .syscfg file once imported into CCS) or add C2000 SysConfig and driverlib support to an existing project.

**Resources:**

- C2000 SysConfig (Software Guide)
- C2000 SysConfig (Application Note)
- SysConfig Development Tool for C2000 real-time MCUs (Video Series)
- Speed Up Development with C2000 Real-Time MCUs Using SysConfig (White Paper)
- *How do I add SYSCONFIG support (Pinmux and Peripheral Initialization) to an already existing driverlib project* has information on adding SysConfig support to an existing software project.

## 6.3 Cloud Tools

TI offers a platform of cloud tools found on dev.ti.com. Amongst these various tools is Resource Explorer, which contains development tools, device documentation, and software resources. While these tools are also offered outside TI's cloud tools, dev.ti.com creates a singular tools package page for anyone developing with TI's products.

If a hardware platform is connected to your computer, dev.ti.com will auto detect the device and immediately filter content tailored to that hardware platform as long as TI Cloud Agent is installed. Alternatively, if you do not have the Agent installed, you can manually select the device using the dropdown option below the Get Started section.

## 6.4 Third Party Development Tools

There are several companies who produce C2000 MCU customized interface, simulation (controller, plant, hardware, processor-in-the-loop), and code-generation toolsets for visual, block diagram and model-based design. These products can enable rapid control system prototyping, system modeling, and can be used for debug, test, and conformance, and to speed time to production.

For a full list of third parties, see the C2000 3P Search Tool. Among these the most popular is MathWorks with Embedded Coder. To get started with Mathworks development, see C2000's Embedded Coder Hardware Support Package.

# 7 Support Resources

## 7.1 Documentation Support

The documentation related to a device can be found within the device's page on TI.com/C2000. To access the device's page simply enter the part number into the search bar and click on the part number of interest.

Within the device's page is a Technical Documentation section that contains all relevant documentation to the device. This content can be filtered by literature type or through a keyword search.

The most important documentation for a device is the content listed below:

| | |
|---|---|
| **Data Sheet:** | Provides all of the parameters and functional data information for the particular device. |
| **Technical Reference Manual:** | Details the integration, environment, functional description, and the programming models for each peripheral and subsystem in the device. |
| **Errata:** | Describes known advisories on silicon and provides workarounds. |

In order to receive a notification of documentation updates, click on 'Subscribe to updates' located at the top right of the device's page. This registration enables you to receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.






**下载：**

C2000 SysConfig is delivered through C2000Ware and can be used within CCS。要访问 SysConfig，可以从现有的基于 C2000 SysConfig 的 driverlib 项目开始（这些项目在导入到 CCS 后将包含一个 .syscfg 文件），或者向现有项目添加 C2000 SysConfig 和 driverlib 支持。

**资源：**

- C2000 SysConfig (软件指南)
- C2000 SysConfig (应用说明)
- SysConfig 开发工具 用于 C2000 实时 MCU (视频系列)
- 使用 SysConfig 加速 C2000 实时微控制器 开发 (白皮书)
- 如何将 SYSCONFIG 支持（Pinmux 和 外设初始化）添加到已存在的 driverlib 项目 包含有关向现有软件项目添加 SysConfig 支持的信息。

## 6.3 云工具

德州仪器在 dev.ti.com 上提供了一整套云工具平台。Resource Explorer 是这些工具中的一项，包含开发工具、器件文档和软件资源。尽管这些工具也在 TI 的云工具之外提供，dev.ti.com 为任何使用 TI 产品进行开发的人创建了一个统一的工具包页面。

如果有硬件平台连接到您的计算机，dev.ti.com 会在安装了 TI 云代理 (TI Cloud Agent) 的情况下自动检测设备并立即过滤出针对该硬件平台的定制内容。或者，如果您未安装该代理，您可以使用“入门 (Get Started)”部分下方的下拉选项手动选择设备。

## 6.4 第三方开发工具

有几家公司为 C2000 MCU 生成定制化接口、仿真（控制器、被控对象、硬件、处理器在环）和用于可视化、方块图及模型驱动设计的代码生成工具集。这些产品可实现快速控制系统原型开发、系统建模，并可用于调试、测试和符合性验证，从而加快生产周期。

完整的第三方列表请参见 C2000 3P Search Tool。在这些第三方中，最受欢迎的是 MathWorks 及其 Embedded Coder。要开始进行 MathWorks 开发，请参阅 C2000's Embedded Coder Hardware Support Package。

# 7 支持资源

## 7.1 文档支持

与器件相关的文档可在该器件的 TI.com/C2000 页面中找到。要访问器件页面，只需在搜索栏中输入零件编号并点击感兴趣的零件编号即可。

在器件页面中有一个“技术文档”部分，包含与该器件相关的所有文档。可以按文献类型过滤此内容，或通过关键词搜索进行筛选。

器件最重要的文档如下所列：

| | |
|---|---|
| **数据表：** | 提供特定器件的所有参数和功能数据的信息。 |
| **技术参考手册：** | 详述器件中每个外设和子系统的集成、环境、功能描述和编程模型。 |
| **勘误表：** | 描述已知的硅片通知并提供解决方法。 |

为了接收文档更新通知，请点击器件页面右上角的“Subscribe to updates”。此注册使您能够接收任何已更改产品信息的每周简报。有关更改详情，请查看任何修订文档中包含的Revision History。

## 7.2 Training

To help assist design engineers in taking full advantage of the C2000 microcontroller features and performance, TI has developed a variety of step-by-step training resources. These training resources have been designed to decrease the learning curve, while reducing development time, and accelerating product time to market. A subset of resources is presented below. For a complete list of the various training resources, visit the TI training site.

**Device/IP Level Training:**

- C2000 Academy: The C2000 Academy is a great resource for developers to learn about C2000 real-time microcontrollers platform. The Academy delivers easy-to-use training modules that span a wide range of topics for all C2000 devices. Additionally, C2000 Academy provides hands-on lab exercises and a full solution for each lab exercise.
- Overview Training Videos: These videos contain information about the latest device series, features, key capabilities and peripherals, and foundational safety.

**Control Theory Training:**

- Control Theory Seminar: This is a four-part technical seminar that offers an introduction to control theory covering fundamental concepts, feedback systems, transient response, and discrete-time systems.
- State Space Control Seminar: This is a four-part course in control theory based on the state space modeling paradigm covering state space models, properties of linear systems, state feedback control, and linear state estimators.

**Application Specific Training:**

- Motor Control Training Videos: This collection of videos provides information about motor control, InstaSPIN™, DesignDRIVE, and MathWorks®.
- Digital Power Control Training Videos: These videos cover information about digital power, solar inverter, GaN & SiC based reference designs, and MathWorks.
- Electric Vehicle Training Videos: These videos cover EV specific trainings on inverter and charging applications.

## 7.3 TI E2E Support Forum

An engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need. The Getting Started in E2E page has answers to frequently asked questions for specific tasks in the TI E2E forum in chronological order.

Linked content is provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's Terms of Use.

# 8 References


- Texas Instruments: *The Essential Guide for Developing with C2000 Real-Time Microcontrollers*
- Texas Instruments: *Hardware Design Guide for F2800x C2000™ Real-Time MCU Series*
- Texas Instruments: *How to Maximize GPIO Usage in C2000™ Devices*
- Texas Instruments: *An Introduction to IBIS (I/O Buffer Information Specification) Modeling*
- General information on C2000 real-time MCUs - C2000™ Overview
- C2000 Products - C2000™ Products
- C2000 Design and Development Resources – C2000™ Design & Development







## 7.2 培训

为了帮助设计工程师充分利用 C2000 微控制器 的特性和性能，TI 已开发了多种循序渐进的培训资源。这些培训资源旨在降低学习曲线，同时减少开发时间并加速产品上市。下面呈现了这些资源的一个子集。有关各种培训资源的完整列表，请访问 TI trainingsite.

**器件/知识产权级别培训：**

- C2000 Academy: C2000 学院是开发人员学习 C2000 实时微控制器平台的极佳资源。该学院提供易于使用的培训模块，涵盖所有 C2000 器件的广泛主题。此外，C2000 学院还提供动手实验练习以及每个实验练习的完整解决方案。
- Overview Training Videos: 这些视频包含有关最新器件系列、特性、关键能力和外设以及基础安全的信息。

**控制理论培训：**

- Control Theory Seminar: 这是一个由四部分组成的技术研讨会，介绍控制理论的基础概念、反馈系统、瞬态响应和离散时间系统。
- State Space Control Seminar: 这是一个基于状态空间建模范式的四部分控制理论课程，涵盖状态空间模型、线性系统的性质、状态反馈控制和线性状态估计器。

**Application Specific Training:**

- Motor Control Training Videos: 此系列视频提供有关 motor control、InstaSPIN™、DesignDRIVE 和 MathWorks®。
- Digital Power Control Training Videos: 这些视频涵盖有关 digital power、solar inverter、GaN & SiC 基参考设计以及 MathWorks 的信息。
- Electric Vehicle Training Videos: 这些视频涵盖针对 EV 的逆变器和 charging applications 的专项培训。

## 7.3 TI E2E Support Forum

工程师获取快速、经过验证的答案和设计帮助的首选来源——直接来自专家。搜索现有答案或提出您自己的问题，以获得所需的快速设计帮助。该 Getting Started in E2E 页面按时间顺序列出了 TI E2E forum 中针对特定任务的常见问题解答。

Linked content is provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI的使用条款。

# 8 参考资料


- Texas Instruments: 使用 C2000 实时微控制器开发的基本指南
- Texas Instruments: F2800x C2000TM 实时微控制器系列硬件设计指南
- Texas Instruments: 如何在 C2000TM 器件中最大化 GPIO 使用
- Texas Instruments: IBIS（I/O 缓冲区信息规范）建模简介
- 关于 C2000 实时微控制器的一般信息 - C2000TM 概览
- C2000 产品 - C2000TM 产品
- C2000 设计与开发资源 – C2000TM 设计 & 开发

## Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.





**修订历史**：先前修订版的页码可能与当前版本的页码不同。

## IMPORTANT NOTICE AND DISCLAIMER

TI PROVIDES TECHNICAL AND RELIABILITY DATA (INCLUDING DATA SHEETS), DESIGN RESOURCES (INCLUDING REFERENCE DESIGNS), APPLICATION OR OTHER DESIGN ADVICE, WEB TOOLS, SAFETY INFORMATION, AND OTHER RESOURCES "AS IS" AND WITH ALL FAULTS, AND DISCLAIMS ALL WARRANTIES, EXPRESS AND IMPLIED, INCLUDING WITHOUT LIMITATION ANY IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE OR NON-INFRINGEMENT OF THIRD PARTY INTELLECTUAL PROPERTY RIGHTS.

These resources are intended for skilled developers designing with TI products. You are solely responsible for (1) selecting the appropriate TI products for your application, (2) designing, validating and testing your application, and (3) ensuring your application meets applicable standards, and any other safety, security, regulatory or other requirements.

These resources are subject to change without notice. TI grants you permission to use these resources only for development of an application that uses the TI products described in the resource. Other reproduction and display of these resources is prohibited. No license is granted to any other TI intellectual property right or to any third party intellectual property right. TI disclaims responsibility for, and you will fully indemnify TI and its representatives against, any claims, damages, costs, losses, and liabilities arising out of your use of these resources.

TI's products are provided subject to TI's Terms of Sale or other applicable terms available either on ti.com or provided in conjunction with such TI products. TI's provision of these resources does not expand or otherwise alter TI's applicable warranties or warranty disclaimers for TI products.

TI objects to and rejects any additional or different terms you may have proposed.

Mailing Address: Texas Instruments, Post Office Box 655303, Dallas, Texas 75265


## 重要声明和免责声明

TI 提供技术和可靠性数据（包括 数据表）、设计资源（包括 参考设计）、应用或其他设计建议、网络工具、安全信息和其他资源"按原样"并带有所有缺陷，并且否认所有明示和暗示的保证，包括但不限于任何 适销性暗示保证、 特定用途适用性 或 不侵犯第三方知识产权 的任何暗示保证。

这些资源的目标用户是使用 TI 产品进行设计的熟练的开发人员。您单独负责 (1) 为您的应用 选择适当的 TI 产品，(2) 设计、验证和 测试 您的应用，和 (3) 确保您的应用符合 适用标准，以及任何其他安全、保安、监管或其他要求。

这些资源如有更改，恕不另行通知。TI 授予您许可，仅将这些资源用于 使用资源中描述的 TI 产品 的应用 开发。禁止其他对这些资源的复制和展示。未授予任何其他 TI 知识产权或任何第三方知识产权的许可。TI 声明不承担责任，且您将 全面赔偿 TI 及其 代表 因您使用这些资源而产生的任何 索赔、损害赔偿、费用、损失和责任。

TI's products are provided subject to 销售条款 or other applicable terms available either on ti.com or provided in conjunction with such TI products. TI' s provision of these resources does not expand or otherwise alter TI' s applicable 保修/保证 or 保修免责声明 for TI products.

TI反对并拒绝您可能提出的任何额外或不同条款。

Mailing Address: 德州仪器, Post Office Box 655303, 德克萨斯州达拉斯市 75265 版权所有